SECTION 33 81 26
COMMUNICATIONS UNDERGROUND DUCTS, MANHOLES, AND HANDHOLES (ALL REGIONS)

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PART 0 PROJECT-SPECIFIC AMENDMENTS

The following amendments to this Section apply only to this Project:

- 2.1.D: Replace full line item with: “To be confirmed by the Design Assist Contractor and approved by the client.”
- 2.2, 2.5, 2.7 & 2.9: Delete this Section in its entirety.2.3.A2: Replace “Oldcastle Infrastructure” with “Suitable manufacturer, to be approved.”
- 2.3.D2: Replace “Oldcastle Infrastructure” with “Manufacturer, to be approved.”

PART 1 GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and the Construction Agreement, including any Amendments thereto, and Sections in Division 01, apply to this Section.

1.2 SUMMARY

A. Nonmetallic underground telecommunications conduit.

B. Telecommunications vault.

C. Accessories.

1.3 RELATED REQUIREMENTS

A. Section 01 25 00 - Substitution Procedures

B. Section 01 33 00 - Submittal Procedures

C. Section 01 60 00 - Product Requirements

D. Section 01 77 00 - Closeout Procedures

E. Section 03 30 00 - Cast-In-Place Concrete

F. Section 07 90 05 - Joint Sealers

G. Division 27 - Telecommunications: All applicable sections

H. Section 31 23 23 - Fill, Backfill and Compaction

I. Section 31 23 16.13 - Trenching For Site Utilities

J. Section 33 41 00 - Subdrainage

K. Section 32 17 23 - Pavement Markings

L. UID 20041

1.4 REFERENCE STANDARDS

A. Referenced standards shall be the latest editions adopted as of the project bid date, except that if an earlier edition is cited in a code or law adopted by the local AHJ, that edition shall take precedence.

B. American Association of State highway and Traffic Officials (AASHTO):
   1. AASHTO LFRD Bridge Design Specifications (including H-20 specification).
   2. AASHTO M 306 - Standard Specification for Drainage, Sewer, Utility, and Related Castings.

C. ASTM International (ASTM):
   1. ASTM C891 - Standard Practice for Installation of Underground Precast Concrete Utility Structures.
   2. ASTM D3035 - Standard Specification for Polyethylene (PE) Plastic Pipe (DR-PR) Based on Controlled Outside Diameter.
   3. ASTM F2160 - Standard Specification for Solid Wall High Density Polyethylene (HDPE) Conduit Based on Controlled Outside Diameter (OD).

D. European Standards (EN):
   1. EN 124 - Gully tops and manhole tops for vehicular and pedestrian areas (including D400 specification).

E. Telecommunications Industry Association (TIA):
   1. TIA/EIA 569 - Telecommunications Pathways and Spaces.

F. Underwriters Laboratories (UL):
   1. UL 651A - High Density Polyethylene (HDPE) Conduit.

G. Malaysia
   1. MCMC MTSFB TC G024:2020 Fixed Network Facilities – In-building and External
   2. MCMC MTSFB TC G025-1:2020 Basic Civil Works – Part 1: General Requirement
   3. MCMC MTSFB TC G025-2:2020 Basic Civil Works – Part 2: Open Trench
   4. MCMC MTSFB TC G025-3:2020 Basic Civil Works – Part 3: Micro Trench
   5. MCMC MTSFB TC G025-4:2020 Basic Civil Works – Part 4: Horizontal Directional Drilling (HDD)

1.5 SUBMITTALS

A. Submit in compliance with requirements of Section 01 33 00 - Submittal Procedures.

B. Product Data: Provide for nonmetallic conduit, manhole accessories, and cable vault.

C. Shop Drawings:
   1. Cable Vault: Submit Shop Drawings indicating size and arrangements for proposed cable vaults, including location and mounting of accessories and connections to adjacent components.
   2. Transition Details: Submit Shop Drawings confirming the arrangement for transition between different OSP conveyance types.

D. IFC Redlines meeting the following Minimum Data Collection Requirements for Underground Telecommunication Record Drawings shall be stamped and submitted for A/E of Record's approval.
   1. Minimum Data Collection Requirements for Underground Telecommunications:
      a. This section shall address duct banks and raceways associated with underground telecommunication lines.
      b. Provide survey grade state plane Global Positioning System (GPS) of Field

collected survey electronic data horizontal and vertical coordinates for each manhole, bend, encasement, conduit count, and so on.

c. Provide horizontal and vertical coordinates along linear duct bank, trench, and trestle installations at an interval not less than once per 100 feet (30 m) of installed conveyance.
d. Include notes for any special construction features such as changes in encasement, conveyance type, sizes, and ratings.

2. Minimum Requirements for Utility Crossings during Telecommunication Installation:
   a. When a new utility installation requires the crossing of an existing utility within the site, the project record drawings shall indicate the location, size, and elevation at which the existing utility was encountered.
   b. Provide survey grade state plane Global Positioning System (GPS) of Field collected survey electronic data horizontal and vertical coordinates for each utility visually identified during the installation of the subject utility.
   c. Temporary utilities can be omitted from this requirement.
3. Redlines shall include noted plans indicating at a minimum the information listed in Items a, b, and c above and shall be accompanied with electronic GPS coordinate file, or survey ASCII file in electronic format. Field notes shall be provided upon request.
4. Records shall be provided in digital formats as outlined in Section 01 77 00 - Closeout Procedures.

E. Project Record Documents: Record actual routing and elevations or depths of underground conduit and duct, and locations and sizes of manholes. Include notes for any special construction features such as changes in depths, slopes, encasement, trenches, and changes in conduit types and sizes. See Section 01 78 39 - Project Record Documents.

1.6 QUALITY ASSURANCE

A. Manufacturer Qualifications: Company specializing in manufacturing the products specified in this section with minimum three (3) years documented experience.

PART 2 PRODUCTS

2.1 UNDERGROUND TELECOMMUNICATIONS CONDUIT

A. Description: Rigid Plastic Underground Conduit: Inside diameter of 2 inch (50 mm), with minimum pull tensile strength of 1500 pounds (680 kg), ASTM D3035.

B. Material: High-Density Polyethylene (HPDE) SDR 11; external smooth finish, internal longitudinal ribs. Schedule 80 HDPE is allowed as a substitute only where local codes require UL listed telecommunications conduit. Color shall be compliant with local utility code requirements, or orange if not specified.

C. Compliant with cell classification material requirement specified in ASTM F2160.

D. Manufacturer and product or model:

1. To be confirmed by the Design Assist Contractor and approved by the client.

~~1. Innerduct (www.innerduct.com)~~
~~2. Duraline (www.duraline.com)~~
~~3. Carlon (www.carlonsales.com)~~
~~4. Weterings Plastics (www.weteringsplastics.nl)~~
~~5. EFLEX (Japan).~~

E. Substitutions: See Section 01 25 00 - Substitution Procedures.

~~2.2 CABLE TRENCH AND ACCESSORIES (US ONLY)~~

~~A. Description:~~
~~1.Plastibeton surface trench as manufactured by Oldcastle and similar to Civil details.~~

~~B. Material:~~
~~1.Non-porous, dielectric, and resistant to degradation from extreme temperature changes, oil, acid, grease, or salt, and AASHTO H-20 rated (EN124 D400).~~

~~C. Drainage pipework shall be compliant with utility specifications and coordinated with storm drainage utility plan.~~

~~D. Manufacturer and product or model:~~
~~1.Oldcastle Infrastructure~~
- ~~a. Trench Duct~~
  - ~~1) T4016 HDPC Trench~~
- ~~b. Accessories~~
  - ~~1) Locking and Non-Locking Covers~~
    - ~~a) Locking covers provided every twenty feet (20 feet [6 m]).~~
    - ~~b) Non-locking covers provided between locking covers.~~
    - ~~c) A minimum of 2 locking covers between FTBs and straight sections.~~
    - ~~d) Composite covers with pick holes:~~
      - ~~i. Provided in non-vehicular traffic areas.~~
    - ~~e) Steel covers with pick holes:~~
      - ~~i. Provided in vehicular traffic or parking areas.~~
  - ~~2) Lock~~
    - ~~a) Penta lock and key~~
      - ~~i. Turn over key to Owner at Project completion.~~
  - ~~3) Lifting Tool~~
    - ~~a) 34655212~~
    - ~~b) Turn over lifting tool to Owner at Project completion.~~
  - ~~4) End Caps~~
    - ~~a) #4016 End Plate~~
      - ~~i. 33690012~~
    - ~~b) Manufacturer provided endcaps shall be installed where trench product deadends with no connection.~~
  - ~~5) Mule Tape~~
    - ~~a) Shall be installed the length of the trench run through FTBs and other junctions to allow for cable installation.~~

~~E. Reference:~~
~~1.Civil Details in Contract Drawings.~~
~~2.UID 20041~~

~~F. Substitutions:~~
~~1.See Section 01 25 00 - Substitution Procedures.~~

2.3 CABLE TRENCH AND ACCESSORIES (NON-US AT POE ENTRANCES)

A. Description:
1. Contractor shall cast in place trench to the trench duct dimensions provided and as shown in the civil details.
2. Provide trench lip from ~~Oldcastle Infrastructure~~ suitable manufacturer, to be approved.

B. Material:
1. Cast in place concrete.

C. Drainage pipework to be compliant with utility specifications and coordinated with storm drainage utility plan.

D. Product:
1. Trench Duct (cast in place)
    a. Approximate Overall Depth
        1) 50.64 cm
    b. Approximate Overall Width
        1) 115.88 cm
    c. Approximate Internal Depth
        1) 39 cm
    d. Approximate Internal Width
        1) 102 cm
    e. Drainage Openings in Bottom
        1) 25 cm by 38 cm
            a) Two per 3 m section
~~2. Oldcastle Infrastructure~~ Manufacturer, to be approved.
    a. Locking and Non-Locking Covers
        1) Locking covers provided every six meters (6 m).
        2) Non-locking covers provided between locking covers
        3) A minimum of 2 locking covers between FTBs and straight sections
        4) Composite covers with pick holes
            a) Provided in non vehicular traffic areas.
        5) Steel covers with pick holes
            a) Provided in vehicular traffic or parking areas.
    b. Lock
        1) Lock and key
            a) Turn over key to Owner at project completion.
    c. Lifting Tool
        1) Turn over lifting tool to Owner at project completion.
3. Accessories
    a. Endcaps
        1) Manufacturer provided endcaps shall be installed where trench product deadends with no connection.
    b. Mule Tape
        1) Shall be installed the length of the trench run through FTBs and other junctions to allow for cable installation.

E. Reference
1. Civil Details in Contract Drawings.
2. UID 20041.

2.4 CABLE TRENCH AND ACCESSORIES (NON-US)

A. Description:
   1. Precast concrete surface trench similar to Civil details.

B. Material:
   1. Non-porous, dielectric and resistant to degradation from extreme temperature changes, oil, acid, grease, or salt, and AASHTO H-20 rated (EN124 D400)

C. Covers shall be dielectric and rated for the required loading at the installation location, up to and including EN124 D400.

D. Drainage pipework to be compliant with utility specifications and coordinated with storm drainage utility plan.

E. Product:
   1. Trench Duct
      a. Approximate Overall Depth
         1) 50.64 cm.
      b. Approximate Overall Width
         1) 115.88 cm.
      c. Internal Depth
         1) 39 cm.
      d. Internal Width
         1) 102 cm.
      e. Drainage Openings in Bottom
         1) 25 cm by 38 cm.
            a) Two per 3 m section.
   2. Accessories
      a. Locking and Non-Locking Covers
         1) Locking covers provided every six meters (6 m).
         2) Non-locking covers provided between locking covers.
         3) A minimum of 2 locking covers between FTBs and straight sections.
         4) Composite covers with lifting holes
            a) Provided in non vehicular traffic areas.
         5) Steel covers with lifting holes
            a) Provided in vehicular traffic or parking areas.
      b. Lock
         1) Lock and key
            a) Turn over key to Owner at Project completion.
      c. Lifting Tool
         1) Turn over lifting tool to Owner at Project completion.
      d. Endcaps
         1) Manufacturer provided endcaps shall be installed where trench product deadends with no connection.
      e. Mule Tape
         1) Shall be installed the length of the trench run through FTBs and other junctions to allow for cable installation.

F. Reference
   1. Civil Details in Contract Drawings.
   2. UID 20041.

~~2.5 FIBER TRENCH BOXES (FTB) (US ONLY)~~

~~A. Manufacturer and product or model:~~
~~1.Oldcastle Infrastructure~~
~~a. Trench Box~~
~~1) 6 foot by 8 foot by 1 foot-9 inch Fiber Trench Box (FTB) with 5 inch walls~~
~~a) Custom Product Made to Order~~
~~b. Accessories~~
~~1) Lifters~~
~~a) Oldcastle Infrastructure~~
~~i. Four (4) - 6UA444 #2 Lifters at 90 degrees~~
~~2) Cover Lid~~
~~a) INWESCO~~
~~i. 7296ST-BIQP-300L~~
~~3) Spares:~~
~~a) 2 lift assist springs~~

~~B. Reference~~
~~1.Civil Details in Contract Drawings.~~
~~2.UID 20041.~~

2.6 FIBER TRENCH BOXES (FTB) (NON-US)

A. Product:
1.Trench Box
a. 1.8 m by 2.4 m - 229 mm Fiber Trench Box (FTB) with 127 mm walls
1) Custom Product Made to Order. Refer to Civil details for product details.
2.Accessories
a. Lifters
1) Four (4) - 6UA444 #2 Lifters at 90 degrees
b. Cover Lid
1) Design features:
a) The cover shall have (4) doors that are shock assist and will only require about 15-21 kg of lifting force to open.
i. Shock shall be constructed of a galvanized spring and steel rod.
b) The center support tube shall be removable.
c) The cover shall be rated for 1465 kg/m^2 (Pedestrian Rated).
d) Safety hinges with removable locking pins.
e) The cover shall bolt onto any existing or new Fiber Trench Box.
f) Covers have recessed padlock latches for added security and do not create a tripping hazard while locked.
g) The cover shall sit on top of the walls so it shall be 109.5 mm taller than the trench.
2) Finishes:
a) The covers can be hot dipped galvanized.
c. Spares:
1) 2 lift assist springs

B. Reference
1.Civil Details in Contract Drawings.
2.UID 20041.

~~2.7 TELECOMMUNICATIONS UNDERGROUND VAULTS (US ONLY)~~

A. ~~Description: Fiber-reinforced concrete vault with composite vault lid, and sealed ports for conduit entry and exit.~~
   1. ~~Concrete Vault:~~
      a. ~~Size: Capable of accommodating up to 96 by 2 inch internal diameter conduit entries per wall. See Contract Drawings.~~
      b. ~~Concrete Strength: 5800 PSI.~~
      c. ~~Fiber: Polypropylene fiber by 3M Corporation.~~
      d. ~~Concrete Additive for Waterproofing: Xypex Admix, C Series.~~
      e. ~~Conduit Entry Provisions: Custom duct seals. Number of ports as specified by Owner.~~
      f. ~~Conduit Entry: SDR11 HDPE duct entrances manufactured for the purpose.~~
      g. ~~Conduit entry locations in vault walls will facilitate cable routing and slack loops within vault.~~
      h. ~~Each chamber shall have a 12 inches by 12 inches by 6 inches deep sump, located in one corner of the vault, as detailed in the C-838 Contract Drawings.~~
      i. ~~Vault shall be equipped with non metallic fiber slack management equipment, sized and rated for fully populated vault loading, below-grade rated.~~
      j. ~~Manhole/handhole covers are not required to be grounded, except where required by local code.~~
   2. ~~Vault Lid:~~
      a. ~~Opening Size: 36 inch (900 mm) inside clear diameter minimum, located as shown in the C-838 Contract Drawings.~~
      b. ~~AASHTO H20 (EN 124 D400) Load rating for incidental passage. (AASHTO H20 wheel loads definition: 16000 pounds (7257 kg) distributed over a 10 by 20 inch (254 by 508 mm) area with 30 percent impact factor).~~
      c. ~~Patented lock assembly located outside the main chamber in an appendage to prevent unauthorized entry and moisture infiltration.~~
      d. ~~GRP material preferred.~~
   3. ~~Manufacturer and product or model:~~
      a. ~~Fiberlite (www.fiberlite.com)~~
      b. ~~Neenah Foundry (www.neenahfoundry.com)~~
   4. ~~Vault Accessories:~~
      a. ~~Conduit Entry and Exit Ports:~~
         1) ~~Blank Duct Plugs: Sealing devices for Ducted Cable Networks, watertight duct plugs.~~
         2) ~~Fiber Optic Simplex Plugs: Sealing plugs for small diameter ducts, watertight duct plugs. Review Contract Drawings and confirm with Owner the plug diameters to be used.~~
         3) ~~Manufacturer and product or model:~~
            a) ~~TE (www.te.com)~~
            b) ~~Roxtec (www.roxtec.com)~~
      b. ~~Fiber Storage Unit (FSU):~~
         1) ~~20 inch (508 mm) diameter fiber storage units for fiber-optic cable.~~
         2) ~~Manufacturer; Model: AFL (www.aflglobal.com); FOSP-20-ADSS-TP or model as suited for installation.~~
      c. ~~Cable Slack Management Equipment: to allow for fiber/cable slack management inside the vault.~~
         1) ~~Manufacturer / Model: Underground Devices Inc.(www.udevices.com); series CRxx-B stanchion, RA20LP [contact: wlewis@udevices.com, or~~

~~jcopeland@udevices.com].~~

~~2) Manufacturer and product or model:~~

~~a) Struttech (struttech.com)~~

~~i. RA2000, series 202 channel~~

~~d. All hardware as required, corrosion-resistant and rated for below-grade installations.~~

~~B. Reference~~

~~1. Civil Details in Contract Drawings.~~

~~2. UID 20041.~~

2.8 TELECOMMUNICATIONS UNDERGROUND VAULTS (EMEA ONLY)

A. Description: Fiber-reinforced concrete vault with composite vault lid, and sealed ports for conduit entry and exit.
1. Concrete Vault:
   a. Size: Capable of accommodating up to 2337 by 51 mm internal diameter conduit entries per wall. See Contract Drawings.
   b. Concrete Strength: C32/40 with exposure class DC4.
   c. Fiber: Polypropylene fiber by 3M Corporation.
   d. Concrete Additive for Waterproofing: Xypex Admix, C Series.
   e. Conduit Entry Provisions: Custom duct seals. Number of ports as specified by Owner.
   f. Conduit Entry: SDR11 HDPE duct entrances manufactured for the purpose.
   g. Conduit entry locations in vault walls will facilitate cable routing and slack loops within vault.
   h. Each chamber shall have a 300 mm by 300 mm by 150 mm (12 inch by 12 inch by 6 inch) deep sump, located in one corner of the vault, as detailed in the C-836 Contract Drawings.
   i. Vault shall be equipped with non metallic fiber slack management equipment, sized and rated for fully populated vault loading, below-grade rated.
   j. Manhole/handhole covers are not required to be grounded, except where required by local code.
2. Vault Lids:
   a. Main Vault Opening Size: 900 mm (36 inch) inside clear diameter minimum, located as shown in the C-838 Contract Drawings.
   b. Sump Lid Opening Size: 300 mm (12 inch) inside clear diameter minimum, located as shown in the C-838 Contract Drawings.
   c. AASHTO H-20 (EN 124 D400) Load rating for incidental passage.
   d. Patented lock assembly located outside the main chamber in an appendage to prevent unauthorized entry and moisture infiltration.
   e. GRP material preferred.
   f. Manufacturer:
      1) Fiberlite (www.fiberlite.com)
      2) Neenah Foundry (www.neenahfoundry.com)
3. Vault Accessories:
   a. Conduit Entry and Exit Ports:
      1) Blank Duct Plugs: Sealing devices for Ducted Cable Networks, watertight duct plugs.
      2) Fiber Optic Simplex Plugs: Sealing plugs for small diameter ducts, watertight duct plugs. Review Contract Drawings and confirm with Owner plug diameters to be used.
      3) Manufacturers: TE (www.te.com), Roxtec (www.roxtec.com)

b. Fiber Storage Unit (FSU):
   1) 508 mm (20 inch) diameter fiber storage units for fiber-optic cable.
   2) Manufacturer; Model: AFL (www.aflglobal.com); FOSP-20-ADSS-TP or model as suited for installation.

c. Cable Slack Management Equipment: to allow for fiber/cable slack management inside the vault.
   1) Manufacturer / Model: Underground Devices Inc.(www.udevices.com); series CRxx-B stanchion, RA20LP [contact: micheleruiz@anixter.com].
   2) Manufacturer / Model: Struttech (struttech.com); RA2000, series 202 channel

d. Fiberglass fixed Ladder to be designed and installed in accordance with EN 14396 and manufacturer's recommendations.

e. Hardware as required, corrosion-resistant and rated for below-grade installations.

B. Reference
   1. Civil Details in Contract Drawings.
   2. UID 20041.

~~2.9 TELECOMMUNICATIONS UNDERGROUND VAULTS (APAC)~~

~~A. Description: Vault with composite vault lid, and sealed ports for conduit entry and exit.~~
   ~~1. Concrete Vault:~~
      ~~a. Size: Capable of accommodating up to 2400 by 50 mm (96 by 2 inch) internal diameter conduit entries per wall. See Contract Drawings.~~
      ~~b. Concrete Strength: 40 MPa (5800 psi).~~
      ~~c. Fiber: Polypropylene fiber by 3M Corporation.~~
      ~~d. Concrete Additive for Waterproofing: Xypex Admix, C Series.~~
      ~~e. Conduit Entry Provisions: Custom duct seals. Number of ports as specified by Owner.~~
      ~~f. Conduit Entry: SDR11 HDPE duct entrances manufactured for the purpose.~~
      ~~g. Conduit entry locations in vault walls will facilitate cable routing and slack loops within vault.~~
      ~~h. Each chamber shall have a 300 mm by 300 mm by 150 mm (12 inches by 12 inches by 6 inches) deep sump, located in one corner of the vault, as detailed in the C-838 Contract Drawings.~~
      ~~i. Vault shall be equipped with non metallic fiber slack management equipment, sized and rated for fully populated vault loading, below-grade rated.~~
      ~~j. Manhole/handhole covers are not required to be grounded, except where required by local code.~~
   ~~2. Vault Lid:~~
      ~~a. Opening Size: 900 mm (36 inch) inside clear diameter minimum, located as shown in the C-838 Contract Drawings.~~
      ~~b. AASHTO H-20 (EN 124 D400) Load rating for incidental passage. (AASHTO H-20 wheel loads definition: 12246 kg (27000 pounds) distributed over a 254 by 508 mm (10 by 20 inch) area with 30 percent impact factor).~~
      ~~c. Patented lock assembly located outside the main chamber in an appendage to prevent unauthorized entry and moisture infiltration.~~
      ~~d. GRP material preferred.~~
      ~~e. Malaysia required a rectangular vault lid with a minimum dimension of 990 x 680 mm (39 inches x 27 inches) hinged type.~~
   ~~3. Manufacturer and product or model:~~
      ~~a. Fiberlite (www.fiberlite.com)~~
      ~~b. Neenah Foundry (www.neenahfoundry.com)~~

4. ~~Vault Accessories:~~
   a. ~~Conduit Entry and Exit Ports:~~
      1) ~~Blank Duct Plugs: Sealing devices for Ducted Cable Networks, watertight duct plugs.~~
      2) ~~Fiber Optic Simplex Plugs: Sealing plugs for small diameter ducts, watertight duct plugs. Review Contract Drawings and confirm with Owner plug diameters to be used.~~
      3) ~~Manufacturer and product or model:~~
         a) ~~TE (www.te.com)~~
         b) ~~Roxtec (www.roxtec.com)~~
   b. ~~Fiber Storage Unit (FSU):~~
      1) ~~508 mm (20 inch) diameter fiber storage units for fiber-optic cable.~~
      2) ~~Manufacturer; Model: AFL (www.aflglobal.com); FOSP-20-ADSS-TP or model as suited for installation.~~
   c. ~~Cable Slack Management Equipment: to allow for fiber/cable slack management inside the vault.~~
      1) ~~Manufacturer / Model: Underground Devices Inc.(www.udevices.com); series CRxx-B stanchion, RA20LP [contact: wlewis@udevices.com, or jcopeland@udevices.com].~~
      2) ~~Manufacturer and product or model:~~
         a) ~~Struttech (struttech.com)~~
            i. ~~RA2000, series 202 channel~~
   d. ~~Hardware as required, corrosion-resistant and rated for below-grade installations.~~

B. ~~Reference~~
   1. ~~Civil Details in Contract Drawings.~~
   2. ~~UID 20041.~~

2.10 ACCESSORIES

A. Non-detectable Underground Warning Tape: minimum 3 inch (75 mm) wide plastic tape, colored orange, or color as required by local code for the application, with suitable warning legend in English and the native language describing buried fiber-optic cables along its entire length.
   1. Material: Tape consists of a minimum 4.0 mil (0.1 mm) overall thickness, with a tensile strength of 2750 psi (19 MPa).
   2. Manufacturer; Model: Ideal Industries; www.idealindustries.com; 42-104, or similar.
   3. Malaysia
      a. Color
         1) Green

B. Tracer Wire and terminal block:
   1. 16 AWG (1.5 mm^2) insulated solid copper tracer wire, rated for direct burial.
   2. Terminal block, rated for corrosive environments, sized to accommodate the maximum number of tracer wires per vault.

C. Surface Utility Markers:
   1. Impact-resistant, triangular-shaped plastic marker post.
   2. Markers to carry fade-resistant decal “Warning - Fiberoptic Cable” on all 3 sides.
   3. Markers to carry the “CBYD” phone number, confirm with Owner.
   4. Markers to carry the telecommunications vault identifier, when marker is associated with a telecommunications vault

5. Length: 78 inch (1980 mm).
6. Installation: Directly buried 18 inch (450 mm) deep into soil, secured with anchor barbs.
7. Markers shall be designed to withstand snow push from street cleaning.
8. Manufacturer; Model: Rhino (www.rhinomarkers.com); TriView.

D. Surface Applied Markers:
   1. Manufacturers:
      a. ACP International (www.acpinternational.com/)
      b. Berntsen (www.berntsen.com)

E. Conduit proofing/cleaning materials
   1. Manufacturers: Condux (www.condux.com)

F. Substitutions: See Section 01 25 00 - Substitution Procedures.
G. Reference
   1. Civil Details in Contract Drawings.
   2. UID 20041.

PART 3 EXECUTION

3.1 GENERAL

A. Verify existing underground utilities before commencing work. Verify that field measurements are as indicated in the Contract Documents. Prevent damage to existing facilities.
   1. Before performing trenching a Ground Penetrating Radar (GPR) survey shall be performed by a certified technician.
   2. Survey grade Global Positioning System (GPS) shall be used to provide spatial accuracy of +/- 0.2 ft (0.06 m) for 95 percent of recorded points.
   3. The survey results shall be incorporated into IFC Redline Set.

B. Verify routing and termination locations of conduit banks prior to excavation for rough-in.

C. Verify locations of manholes prior to excavating for installation.

D. Areas affected by the work must be restored to their original condition, or as required on civil and landscaping Contract Drawings, at the end of construction.

E. Install and place in final position any item, equipment, or material complete and ready for the intended use; fully connected and interfaced, and in operable condition per manufacturer recommendations unless otherwise noted; and perform any manufacturer or Owner required testing and documentation.

3.2 UNDERGROUND CONDUIT

A. Excavate for duct bank installation as specified in Section 31 23 16.16 - Structural Excavation For Minor Structures.

B. Install underground conduits at depths as indicated on civil Contract Drawings, at a minimum of 4 feet (1.20 m) below final grade, and at a minimum of 1 foot (0.30 m) horizontally and vertically away from other underground utilities.

C. Install duct with minimum slope of 4 inches per 100 feet (100 mm per 30 m; 0.33 percent).

D. Slope duct away from building entrances.

E. Install in single, uninterrupted runs, without the use of fittings or connections between cable vaults.

F. Install no more than the equivalent of one 90-degree bend between pull points.

G. Switchbacks (S curves) with a minimum of 49 feet (15 m) between curves

H. 90 degree bends shall have at least a radius of 20 times its diameter.

I. Provide suitable fittings to accommodate expansion and deflection where required at the cable vault.

J. Terminate conduit at manhole entries using appropriate Roxtec seals, and extend 3 inches (75 mm) past the inside wall of the cable vault to allow attachment of couplings.

K. Conduits **shall not** be banded together before backfilling approved **spacers** shall be used, coordinate with Owner.

L. Backfill shall be excavated material, or dirt that is free of debris and cobbles greater than 1/2 inch (13 mm) in diameter.

M. Concrete encasement is only required in telecom conduit banks where required by local code and conditions (such as road crossings).

N. Swab conduit. Use suitable caps to protect installed duct against entrance of dirt and moisture.

O. Unused underground conduits shall be cleaned and capped.

P. Underground conduits shall have a 600 lb (270 kg) stretch-tension mule tape. The mule tape shall be tied to its duct cover/plug to keep it from being pulled back into the duct.

Q. New installed underground conduit shall be tested using a mandrel-blowing process after trenches have been backfilled or after a directional bore.

R. Tracer wire shall be installed on the outside of the underground conduits. Provide one tracer wire per fiber route segment, independent of conduit count, on top of and centered relative to conduit segment bundle. Secure it with tape or twists to conduit at 36 inch (1 m) intervals. Tracer wire will enter vault wall, where the penetration will be sealed, or terminate in a terminal post at or close to the cable vault. Provide tracer wire support up to terminal block in vault. Terminate tracer wire on terminal block in vault.

S. Terminal posts must be installed close to the vault cover rim.

T. Telecommunication duct bank conduits shall be integrally colored "orange," or the regional color code for telecommunications systems.

U. Conduits entering building shall have duct plugs and/or simplex plugs to seal conduits and cabling.

V. Contractor shall not add any additional offsets and in particular back to back offsets.

W. Full lengths of the runs from end to end are to be fully photo documented with pictures, to include dedicated pictures of bends, offsets, and penetrations; after they are installed and before being covered.

X. Additional offsets and specifically back to back offsets will adversely impact the pulling of the cables due to their dual longitudinal strength members since it will be impossible to control their orbital rotation during the pull, coordinate with Owner for unplanned or extreme conduit offsets and bends.

Y. In general Duct shall:

1. Run straight with minimal bends.
2. Cross high or medium voltage or any other utilities at a perpendicular angles.
3. Be run above electrical conduits.
4. Be set at a minimum of 4 ft (1.2 m) below grade.
5. Use long radius / swept bends - minimum x20 D.
6. Use concrete encasement under roadways or parking lots etc.
7. Have a maximum height differential from underground pathways to a cable vault entry/intercept of 2 ft (600mm).

3.3 TELECOMMUNICATIONS VAULT AND TRENCH INSTALLATION

A. Excavate for cable trench and/or manhole installation under the provisions of Section 31 23 16.16 - Structural Excavation For Minor Structures.

B. Install and seal precast sections in accordance with ASTM C891, and Section 07 90 05 - Joint Sealers.

C. Verify site vault layout with Contract civil drawings. Cable vaults shall not extend partially or fully into a paved roadway.

D. Install telecommunications vault slightly non-level, as recommended by manufacturer. Install lid as level as possible.

E. Conduit penetrations shall be caulked and sealed.

F. Conduits shall extend into the interior of the vault by 3 inches (75 mm) nominal, to facilitate the use of blow or jet feed, check Contract Drawings.

G. For the conduit numbering and labeling system, comply with Owner's requirements.

H. Backfill manhole excavation as specified in Section 31 23 16.13 - Trenching For Site Utilities and Section 31 23 23 - Fill, Backfill and Compaction.

I. Manhole lids shall be identified with cast 4 inch (100 mm) high letters as follows: TELECOM

J. The Zero Vaults shall be located within 15 feet (5 m) of the outside of the outer security fence but remain within the property line, unless there is insufficient space to accommodate the vaults. Where there is insufficient space, the zero vault shall be located within 15 feet (5 m) inside the outer security fence.

K. Trench sections shall have two lockable lids between FTBs and one per 20 foot (6 m) section.

L. Contractor shall photo document the Vaults per the Cable Vault Photograph Process, coordinate with Owner for additional information.

M. Vaults shall in general:
  1. Be waterproof (including lids and all conduit entries)
  2. Have lockable lids.
  3. Be placed to limit conduit run lengths to between 150 m and 200 m. Coordinate with Owner when exceeding 150 m.
  4. Be used for conduit run direction changes of 22.5 degrees or greater.
  5. Not be place within 15m of fueling areas
  6. Be used primarily as pulling points.

3.4 UNDETECTABLE UNDERGROUND WARNING TAPE

A. Install underground warning tape 12 inch (300 mm) above conduits, colored “orange” or the regional color code for telecommunications systems and marked “TELECOM”.

3.5 SURFACE UTILITY MARKERS

A. Surface utility markers are required where conveyance is underground.

A. Locate markers 300 feet (90 meters) apart maximum on straight runs with clear line of sight. At turns and in landscaped areas, locate as needed to indicate turns and allow visibility between markers.

B. Markers must extend at least 3 feet (0.90 meters) above ground.

C. Markers must be placed within 5 feet (1.50 meters) of the actual route.

D. On concrete or asphalt surfaces, comply with requirements of Section 32 17 23 - Pavement Markings.

END OF SECTION

REVISION HISTORY

Version	Date	Description and location of change
0.1	February 2025	Draft IFC Updates
1.0	12th of March 2025	Final IFC Issue